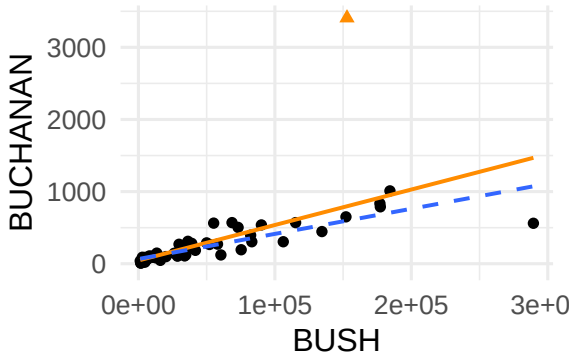
# Day 13: Diagnostics for MLR

With your group: recap the LINE conditions from SLR and how we checked each one. Which of them carry over to MLR, and how would we check them there?

| **LINE condition** | **How we checked it in SLR** | **How we check it in MLR** |
|---|---|---|
| **L**inear | | |
| **I**ndependent errors | | |
| **N**ormally distributed errors | | |
| **E**qual variance of errors | | |

## Regression is not resistant to outliers

Example from CS1. Solid line = fit with outliers included; dashed line = fit with outlier removed. One extreme county visibly "tilts" the line



**Why SLR isn't resistant:**

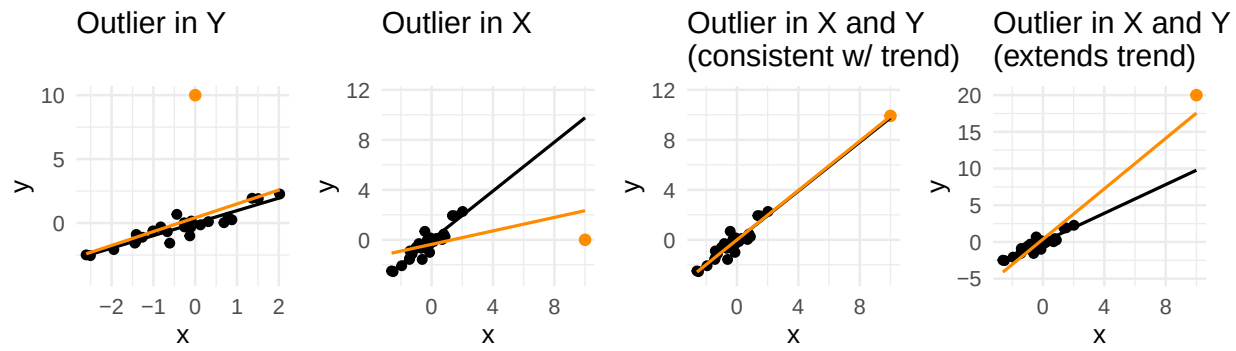
$$\hat{\beta}_1 = r \cdot \frac{s_y}{s_x} \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

**MLR is no better**: we're still minimizing a *sum* of squared residuals and sums aren't resistant to outliers either.

$$SSE = \sum_{i=1}^{n} \left( y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_p x_{ip}) \right)^2$$

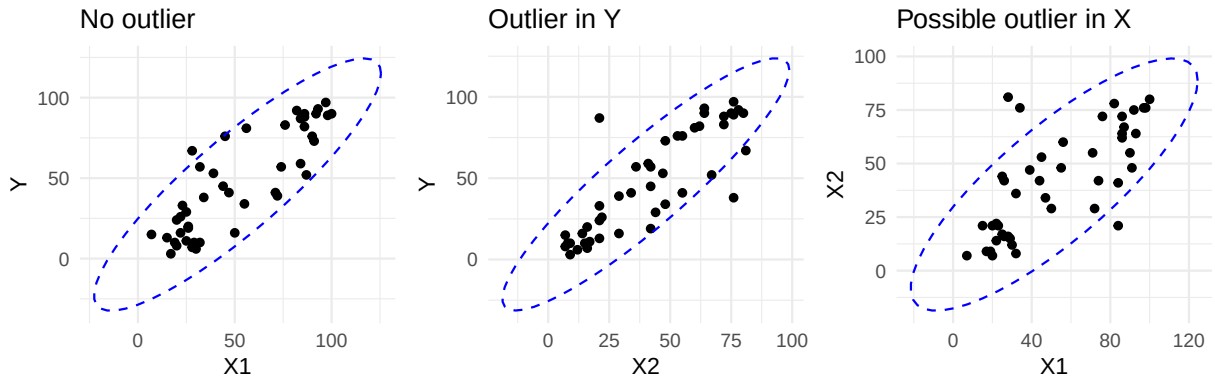
## Types of outliers

An observation can be unusual in $y$, in $x$, or both, and each type pulls the fitted line differently. Below, one added point (orange) plays each role against the same base cloud of points.




Points that are able to substantially change the fitted model are called _____. Not every outlier is influential – an outlier that falls *along* the existing trend barely moves the line (panel 3 above), while one that breaks the trend can swing it a lot (panels 1, 2, 4).

In higher dimensions, outliers can be tricky to spot – detecting an outlier in the predictor space means considering the multivariate relationship *among all* predictors, not just one at a time.



How do we find outliers and decide if they're influential?

- Plot the data
- Plot the standardized residuals
- Fit the model with and without the point(s)
- Calculate *influence diagnostics* – leverage, standardized residuals, DFFITS, Cook's distance

### Leverage (aka "hat")

Leverage measures a point's **potential to influence** the fitted line, based only on _____

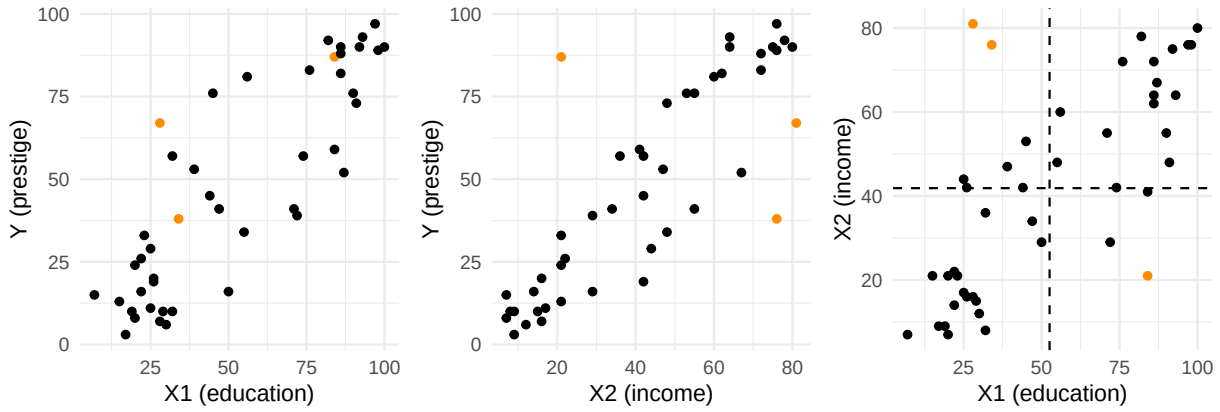
For SLR: $h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum(x_i - \bar{x})^2}$

For MLR: more complicated – it's a distance of a case from the average of all $p$ predictors.

"Flares up" when a point's $x$ is far from the mean, indicating *potential* to be influential (whether it actually is depends on $y$ too).

**Cutoffs:** $2(p+1)/n$ or $3(p+1)/n$

In higher dimensions, leverage is focusing on the distance from the average of all predictors.




### Standardized Residuals

Standardized residuals divide each residual by its standard deviation, which builds in a "penalty" for high-leverage points (a high-leverage point pulls the line toward itself, shrinking its raw residual):

$$r_i = \frac{e_i}{\hat{\sigma}\sqrt{1 - h_i}}$$

"Flares up" when $y_i$ is far from $\hat{y}_i$.

**Cutoffs:** $|r_i| > 2$ for small data sets or $|r_i| > 4$ for large data sets

Cook's distance

**Cook's distance** measures the effect of case $i$ on **all** of the fitted values at once:

$$D_i = \sum_{j=1}^{n} \frac{(\hat{y}_{j(i)} - \hat{y}_j)^2}{(p+1)\hat{\sigma}^2} = \frac{r_i^2}{p+1}\left(\frac{h_i}{1-h_i}\right)$$

**Cutoff:** $D_i$ near or above _____ indicates large influence; also worth comparing the *relative* size of $D_i$ across cases with an index plot.

## Doing this in R: `augment()`

We never calculate these by hand – `broom::augment()` computes them all for us:

- `.hat`: leverage
- `.std.resid`: standardized residuals
- `.cooksd`: Cook's distance

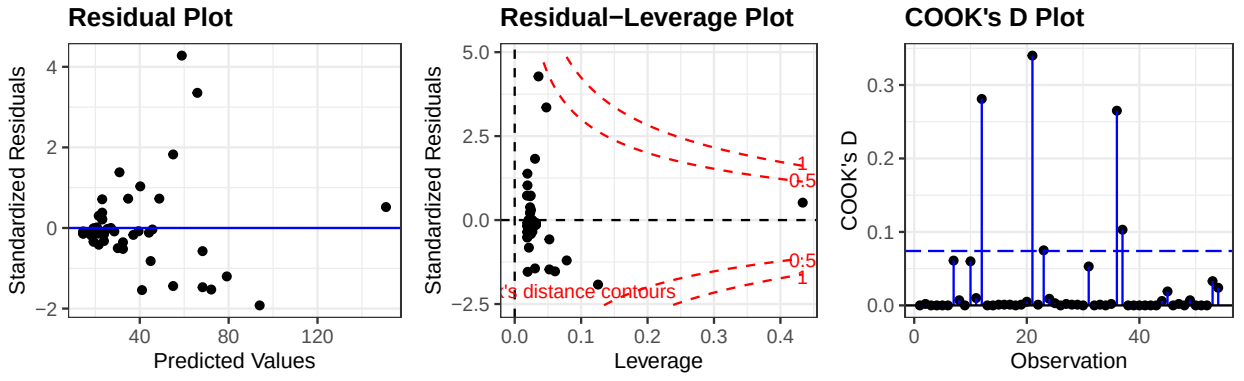
```
augment(bikes_mlr)
```

| rides | temp_actual | .fitted | .resid | .hat | .sigma | .cooksd | .std.resid |
|---|---|---|---|---|---|---|---|
| 654 | 57.400 | 2468.473 | -1814.473 | 0.009 | 1251.355 | 0.005 | -1.455 |
| 1229 | 46.492 | 2208.949 | -979.949 | 0.011 | 1253.251 | 0.002 | -0.786 |
| 1454 | 46.760 | 2232.341 | -778.341 | 0.010 | 1253.540 | 0.001 | -0.625 |
| 1518 | 48.749 | 2405.765 | -887.765 | 0.009 | 1253.392 | 0.001 | -0.712 |
| 1362 | 46.503 | 2209.965 | -847.965 | 0.011 | 1253.448 | 0.001 | -0.680 |
| 891 | 44.177 | 1310.392 | -419.392 | 0.031 | 1253.887 | 0.001 | -0.340 |

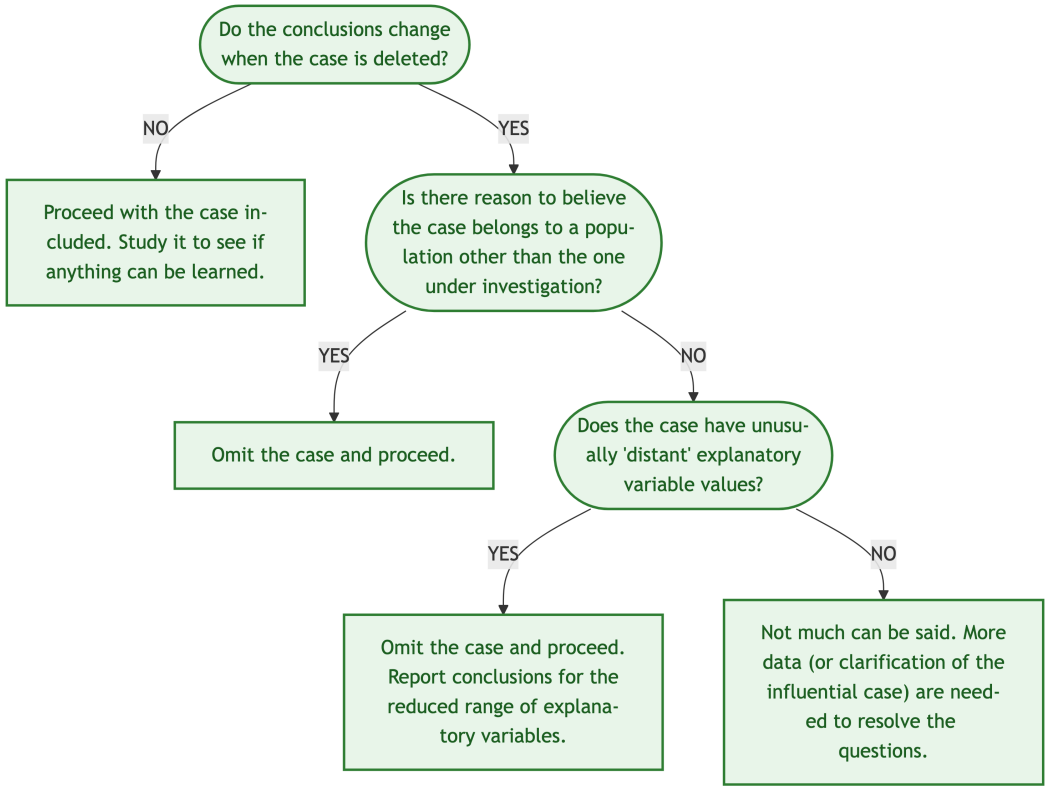
## Shortcut for plots: `resid_panel()`

`ggResidpanel::resid_panel()` gives us residual, leverage, and Cook's distance plots in one call:

```
resid_panel(bigmac.lm, plots = c("resid", "lev", "cookd"), type = "standardized", nrow = 1)
```



What do we do if a case is influential?




*Source: Display 11.8 in* Sleuth *(3rd edition)*